

Bitter melon (*Momordica charantia*) extract suppresses adrenocortical cancer cell proliferation through modulation of the apoptotic pathway, steroid...



Adrenocortical carcinomas are rare but present with extremely poor prognosis. One of the approaches to control cancer progression and reduce cancer risk is prevention through diet. Bitter melon is widely consumed as a vegetable and especially as a traditional medicine in many countries. In this study, we have used human and mouse adrenocortical cancer cells as an in vitro model to assess the efficacy of bitter melon extract (BME) as an anticancer agent. The protein concentrations of BME and other extracts were measured before use. First, BME treatment of adrenocortical cancer cells resulted in a significantly dose-dependent decrease in cell proliferation. However, we did not observe an antiproliferative effect in adrenocortical cancer cells treated with extracts from blueberry, zucchini, and acorn squash. Second, apoptosis of adrenocortical cancer cells was accompanied by increased caspase-3 activation and poly(ADP-ribose) polymerase cleavage. BME treatment enhanced cellular tumor antigen p53, cyclin-dependent kinase inhibitor 1A (also called p21), and cyclic AMP-dependent transcription factor-3 levels and inhibited G1/S-specific cyclin D1, D2, and D3, and mitogen-activated protein kinase 8 (also called Janus kinase) expression, suggesting an additional mechanism involving cell cycle regulation and cell survival. Third, BME treatment decreased the key proteins involved in steroidogenesis in adrenocortical cancer cells. BME treatment decreased the level of phosphorylation of cyclin-dependent kinase 7, which is required, at least in part, for steroidogenic factor 1 activation. Finally, we observed that BME treatment significantly reduced the level of insulin-like growth factor 1 receptor and its downstream signaling pathway as evidenced by lower levels of phosphorylated RAC- α serine/threonine-protein kinase. Taken together, these data illustrate the inhibitory effect of bitter melon on cell proliferation of adrenocortical cancer through modulation of diverse mechanisms.